Development of a BIM-Based Data Management System for Structural Health Monitoring with Application to Modular Buildings: Case Study

Mojtaba Valinejadshoubi[1]; Ashutosh Bagchi[2]; and Osama Moselhi[3]

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Abstract: Modular buildings or off-site construction of building units are increasingly gaining momentum. Although such construction practices have advantages in terms of cost competitiveness and delivery time, they have many issues related to structural integrity and secondary stresses from vibration during transit and misalignment during installation. Therefore, monitoring the vibration, strain, and deformation of the modules using structural health monitoring (SHM) techniques is important. The primary purpose of this study is to explore building information modeling (BIM) techniques to facilitate effective data management and the representation of sensory components of the SHM system in a building and to render or visualize the damage or distress in building components based on the interpretation of sensor data. The proposed framework consists of two main modules: (1) an automated sensor-based data acquisition and storage module, which extracts sensor data for a structure from a corresponding relational database; and (2) an automated data and damage visualization module, through which sensor data are interpreted to identify damage or anomalies in the structure and the affected building components are highlighted and tagged in the BIM of the building to facilitate visualization. The damaged or near-damaged elements of the modules are highlighted in the BIM model through color-coding based on predefined threshold strain values. Because detecting buckled or yielded steel members (local damages) in a building or a module is challenging given that these components are often hidden behind fireproof coating and drywall, the proposed SHM-based condition assessment system will contribute—especially in the preinstallation and operational phases—to providing efficient, near-real-time health monitoring of buildings and increasing the efficiency of the structural condition assessment process. These benefits could be particularly useful for modular buildings, for which the modules are constructed in a plant and transported to the site for installation. In these stages, a module may undergo hidden or visible damage, the installed sensors are expected to provide a mechanism to assess such damage, and the entire process can be managed through BIM. Importantly, note that although a similar concept was explored by other researchers to integrate SHM with BIM, the present study provides a more comprehensive methodology through the complete implementation of the system to demonstrate the concept through a case study. **DOI: 10.1061/(ASCE)CP.1943-5487.0000826.**
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Introduction

The modular construction process represents the highest degree of industrialization of the building construction process, which is currently growing rapidly. Presently, in some construction projects, prefabrication/modular construction is approximately 85% (McGraw-Hill Construction 2011). Clients' requirements for rapid construction, improved quality, and early investment returns are some of the motivating factors for modular construction.

Modular steel buildings are usually composed of prefabricated cold-formed steel assemblies or modules that are fabricated in a manufacturing plant and then shipped to the construction site to be installed to form a building. Compared with hot-rolled steel structures, cold-formed light steel structures are lighter but more susceptible to structural damage given normal and extreme loads (Yang and Bai 2017). A sufficient understanding of the structural behavior of multistory modular buildings subjected to different load types is lacking (Ramaji and Memari 2013). Different situations exist that could lead to the failure of a module, such as increased damage during erection or transportation. Geometric variability is inevitable and can cause problems in the assembly process. A module's component geometry can change from its original design because of problems arising from the manner in which it is handled in the plant, during transportation, and at installation (Rausch et al. 2017). The geometric change may lead exceed the tolerance of fit and generate secondary stresses at installation, causing further damage.

From the manufacturing to the operational phase, modules are subjected to different types of direct and indirect loads. The modules are required to be designed to withstand fabrication, transportation, and installation loads (Naqvi et al. 2014). Predicting the final strength and capacity of a modular building after being transported is difficult, and lifting-induced stresses are not entirely predictable

[1]Ph.D. Student, Dept. of Building, Civil and Environmental Engineering, Concordia Univ., 1455 de Maisonneuve Blvd. West, Montreal, QC, Canada H3G 1M8 (corresponding author). Email: Mojtaba_vlj256@yahoo.com

[2]Professor, Dept. of Building, Civil and Environmental Engineering, Concordia Univ., 1455 de Maisonneuve Blvd. West, Montreal, QC, Canada H3G 1M8.

[3]Professor, Dept. of Building, Civil and Environmental Engineering, Concordia Univ., 1455 de Maisonneuve Blvd. West, Montreal, QC, Canada H3G 1M8.

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(Cantu 2011). Because the structural elements of a module of a building are usually hidden behind the fireproof coatings or drywall, detecting the buckled or yielded steel members (local damages) in a module is challenging (Zhang and Bai 2015). In this context, a useful monitoring tool such as SHM is needed for early structural condition assessment and damage detection in specific parts of the modules in each phase. SHM systems are ranked as one or a combination of the following SHM categories: sensor deployment studies (category 1), anomaly detection (category 2), model validation (category 3), threshold check (category 4), and damage detection (category 5), (Vardanega et al. 2016). Higher-level categories (e.g., categories 4 and 5) have the potential to yield significant values to many stakeholders. According to Webb (2014) and Webb et al. (2014), most published SHM studies are devoted to categories 1 to 3 and the least to category 4.

In practice, monitoring the strain response contributes to an assessment of the structural condition (Park et al. 2013; Ni et al. 2010, 2008). Continuous monitoring of a real-time dynamic strain in a structure can provide valuable information for damage assessments, inspections, and decision making. Strain provides information about the local behavior of structural components and is one of the most used parameters in SHM. Strain is essential in condition monitoring of modular buildings, which can aid in the assessment of damage in structural members in building modules at different stages, and helps assess the reliability of structural components.

Fast and accurate assessment of the structural condition of modules and buildings (modular or other) is essential for timely maintenance and repair to avoid project delays, and is important for occupant safety and occupancy after extreme load events. The challenge, in general, is an effective visualization tool utilizing BIM to make the SHM information easily accessible, understood, and applicable (Zhang and Bai 2015), which could increase the efficiency of the structural condition assessment process and facilities management. BIM can be used effectively to capture the real-time building information that can be used for owners and facility managers to provide accurate and upgraded details on the state of various parts of the building (Chen et al. 2014). BIM combined with real-time monitoring of structural health and damage assessment methods could provide a robust and intelligent system for managing modern buildings, including the modular building type (Seam et al. 2013). However, integrating SHM into the BIM environment has challenges. The study run by Rio et al. (2013) revealed that accomplishing a dynamic monitoring system for the structural behavior of a building to provide sensor data to BIM is not part of BIM functionality and is still a challenge. They concluded that BIM standards need to be extended to allow them to represent monitoring-related information. The study by Wang et al. (2017) found that applying BIM in SHM can improve the effectiveness of monitoring processes and decision making in construction informatics applications.

Despite its potential benefits, few attempts have been made to integrate BIM into SHM. Sternal and Dragos (2016) proposed BIM-based modeling of wireless SHM systems using the industry foundation class (IFC) standard. Although they believed that integrating monitoring-related information into BIM helps categorize, document, and update this information throughout the entire life cycle of the monitored structure, it was mentioned at the end that the current IFC standard does not provide sufficient entities to holistically model and digitally represent an overall wireless SHM system. Theiler et al. (2017) attempted to design a BIM-based prototype SHM and control system by using the extended IFC schema. Smarsly and Tauscher (2016) proposed a conceptual monitoring information modeling built on the IFC standard. Although the authors defined a semantic model to extend the existing IFC 4 standard for digital representation of monitoring-related information, the paper was conceptual and did not include implementation or validation. Del Grosso et al. (2017) attempted to explore the idea of integrating 6D digital models with SHM systems, but the study mainly focused on creating and modeling the sensor system in the BIM software application and did not provide the linkage to sensor data or facilitate visualization. Although the authors of this study discussed the state of the art of the current and potential relationship between SHM and BIM, they believed that the topic is not explored well and needs further investigation. Additionally, it was noted that no preferred workflow exists to inform the BIM model with SHM and interpreted data regarding elements of the project. A preliminary scheme for utilizing BIM to manage SHM data for buildings was developed in Valinejadshoubi et al. (2018c, 2017). BIM was also effectively used in thermal comfort monitoring (Valinejadshoubi et al. 2018b) and the assessment of the seismic risk of nonstructural components in buildings (Valinejadshoubi et al. 2018a).

Although BIM is desired as a dynamic workbench for managing all data related to a building project, connectivity between BIM and SHM is lacking. The full integration between virtual and physical sensors, connecting and inserting sensor data remotely into an external database through Internet of Thing (IoT) technology, and applying a three-dimensional (3D) visualization-based alarm system for SHM projects have not been adequately addressed in previous studies, and the BIM approach has not yet been fully applied and validated for SHM purposes. To address some of these issues, in the present study, a novel integrated system is developed for structural condition monitoring of building components. SHM data are stored in the database and automatically accessed by the BIM model, and the conditions of the relevant building elements are calculated and mapped on the BIM model to visualize the overall state of the structure. The main purpose of this study is to create a mechanism for a BIM model of a building or module to represent and access sensor data, run a data interpretation or damage assessment process, and map it on the corresponding building components. Doing so would facilitate an effective visualization capability for a rapid and efficient structural condition assessment based on the SHM data. The proposed framework can be applied by engineers and facility managers to interpret and assess the ongoing condition of modules during the transportation, installation, and operational phases, to identify hidden damages, and to replace the damaged parts by providing and managing updated monitoring data in a rapid manner to promote timely repair.

BIM and Its Role in Modular Building Construction

Presently, the architectural, engineering, and construction (AEC) industries have been seeking an effective tool for reducing the cost and completion time of projects and increasing their productivity and quality (Azhar et al. 2008). BIM has significantly altered the way that building information is managed by the AEC industry. BIM incorporates digital modeling software to design and manage a project more efficiently (Nassar 2010) and provides powerful new value to construction firms. BIM breaks down the barriers between disciplines by encouraging the sharing of knowledge throughout a project's lifecycle. BIM improves constructability and shortens a project's completion time. In a BIM project, multiple documents are not used in traditional ways; instead, they are digitized and added to a database in BIM software. All information is built into an intelligent BIM model instead of needing to look at separate drawings, schedules, and specifications for the information on a particular element.

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BIM is an organized collection of building data. Regarding BIM, everything begins with a 3D building model. This model has more than just simple geometry added to it for visualization. A true BIM model consists of the virtual equivalent of actual building sections. These intelligent elements are the digital prototype of the physical elements, including walls, columns, windows, doors, specialty equipment, and others. The model allows us to simulate the building and understand its behavior before actual construction begins. Of course, the most basic BIM model is used to create realistic visualizations of the planned building. As previously mentioned, the data in a BIM model are not only used during the design and construction phases of the building project but also throughout the building's life cycle. The building-related data can be easily archived in the BIM model for such things as future usage, analysis, retrieval, and maintenance.

Because modular construction has an additional manufacturing stage relative to conventional construction, utilizing BIM as a powerful information management tool is required. A current issue of offsite construction has been perceived as a process lacking flexibility in design. BIM can partially address this limitation by providing access to a vastly broader range of constituent parts in various levels of detail, from the micro level of an individual fastener to the macro level of a volumetric component (Patlakas et al. 2015). BIM can be used for proper information exchange between different disciplines, which is a fundamental need in multidisciplinary projects such as modular building projects. Data-rich models such as BIM can be used effectively by other members of the design team to coordinate the fabrication of a building's different systems (Nawari 2012). BIM can be used in effective simulation and visualization of a building and its components in digital forms, which are useful for accurate planning of onsite module installations and can resolve any spatial constraints (Han et al. 2011). Due to the large number of elements in modular structures, an automated system is needed for visualizing and monitoring the structural condition of elements in modules in each phase of the project.

SHM in Modular Buildings

A modular building is erected by assembling individual modules. Every module is a primary rectangular frame made of steel or wood frames. The modular units are stacked on top of and or next to one another through a connector at every joint. Two types of connections—vertical (column to column) and horizontal—are used in modular building construction.

Potential failure modes in modular buildings include structural component failures within a modular frame and connection failures at the modular joints. Regarding component failures, corner posts buckling or load-bearing studs and bending of columns/beams may occur. Regarding connection failures at the joints, modular units become separated, which reduces their axial stiffness to zero, causing a critical situation concerning the overall integrity of the entire system. Local connection failure (LCF) occurs due to excessive concentrated force on the connection region, causing complications in the load transfer path. Different types of loads, such as manufacturing, transportation, installation, and operation, are applied to the modules. Any damage to the modules before installation may affect the operational performance of the same. Hence, the modules must be erected without any hidden damage. SHM can be applied for early and rapid structural condition assessment and damage identification of building modules during every phase to assist engineers in deciding on rehabilitation measures when the module's components experience unexpected changes in excessive deformation, deflection, and strain.

SHM system design is developed based on failure modes. It is advantageous for modules' components to be equipped with SHM systems as they are being manufactured. Particular care should be taken regarding the installed sensors on the elements from roughness-induced vibration forces during transportation. Different types of sensors can be applied in the SHM of the modules. For instance, linear variable displacement transducer (LVDT)/ultrasonic sensors are used to measure critical structural deformation (serviceability) and the module separation at the connection region (LCF). Moreover, strain gauges are used to measure the real-time strain on critical structural members in the modular system.

Determining the Locations of Strain Sensors

Modules can be made from light gauge/cold-formed steel or hot-rolled steel such as pipe chassis. However, compared with hot-rolled steel modules, cold-formed light steel modules are more susceptible to deformation and buckling due to applied loads.

The level of strain in structural elements serves as a significant indicator of the level of deformation and damage in the structural and nonstructural elements. Considering this fact, SHM systems often employ strain gauges to measure strains in critical components. Modular construction projects are more complex than conventional ones given the additional manufacturing and transportation processes. Therefore, in addition to operational load, the building modules are subjected to manufacturing, transportation, and installation loads that make their structural elements more susceptible to damage or excessive deformations. For instance, transportation and handling of modules is an important part of the overall life of the modules when they experience high mechanical loads. To identify vulnerable and critical elements, a detailed and accurate structural analysis is needed for each phase for typical modules.

For example, the force from vibrations during the transportation of modules can be simulated in a finite element (FE) model of individual modules to identify their critical elements that could be damaged given road-induced vibration. For this purpose, random vibration data can be used to simulate field and transportation conditions. Random vibration is typically described by power spectral density (PSD) curves of average acceleration intensity in the frequency domain. Different transport vehicles can be related to different PSD shapes and amplitudes. In North America, modules are commonly transported by road (using tractor-trailers). The ASTM D4169 Truck Profile [ASTM D4169 (ASTM 2016)] is among the most widely used road-induced vibration profiles for general simulation and random vibration tests in laboratory experiments. In such cases, the vibration test on a truck is performed at a single intensity level (e.g., Assurance Level II) for the entire test duration. The recently updated version of the standard [ASTM D4169 (ASTM 2016)] recommends the use of three different intensity levels: low, medium, and high, corresponding to the 90th, 95th, and 99th percentile intensities. Fig. 1 shows the PSD levels for different frequencies according to the ASTM D4169 standard, which is typically used for performance testing of shipping containers and systems. The typical PSD units are acceleration [G^2/Hz] versus frequency [Hz]. Note that the amplitude is actually [GRMS^2/Hz], where RMS = root-mean square. The RMS notation is typically omitted for brevity. GRMS is used to define the overall energy or acceleration level of random vibrations. In Fig. 1, profiles 1 to 3 correspond to low-, medium-, and high-level PSD levels, respectively.

As an example, the FE model of an industrial pipe chassis module (hot-rolled steel module) used by Shahtaheri et al. (2017) has been developed here using Autodesk's Simulation Mechanical.

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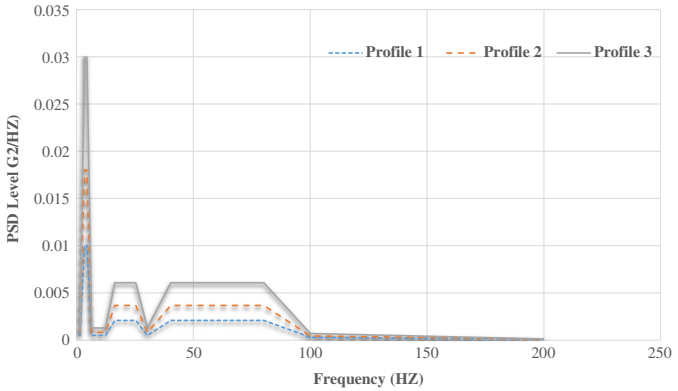


Fig. 1. Random vibration profiles of trucks (ASTM D4169 Truck Assurance Levels 1 to 3).

To simulate the transportation vibration forces, high PSD level, and its corresponding frequencies as given in Fig. 1 were applied to the developed FE model. Fig. 2 shows the FE model of a pipe chassis module. The arrangement of the temporary supports and module dimensions are assumed to be the same as that used in Shahtaheri et al. (2017). As shown in Fig. 2, the most critical elements with the highest displacement should be monitored during the transportation phase. Although the maximum displacement obtained from the analysis is found to be very small (approximately 0.26 mm), in the case of a light gauge/cold formed steel module, it could be much larger. In contrast, uncertainties such as temporary support configuration, which is usually based on experiments rather than objective analysis, might even increase the magnitude of the maximum displacement in the structural elements of the module.

Similarly, the structural behavior of modules could be analyzed against manufacturing, installation, and operational loads to determine their critical elements in each phase.

Data Management Challenges in SHM Process

One of the inherent challenges in SHM is high-volume data management. Management of all original raw data and all post-processed data during the entire life cycle of the structure with a possible size of hundreds of gigabytes can become a problem if all of the original data are to be kept for future processing (Rio et al. 2013). In periodic SHM installations, all data are stored onsite on a data acquisition (DAQ) unit. Additional computation and storage resources are required for continuous monitoring. In most cases,

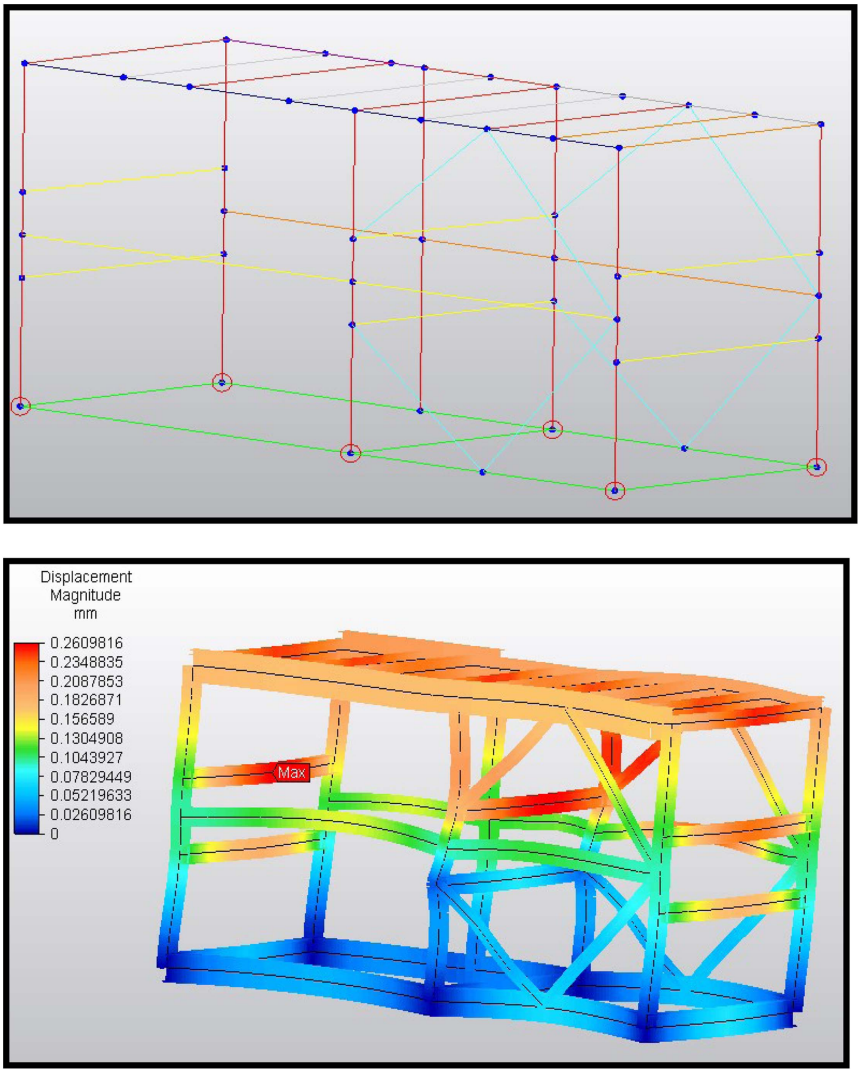


Fig. 2. Sample of random vibration simulation on FE model of pipe chassis module.

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sensors are set at a frequency of 100 Hz and a 16- or 24-bit resolution per sample, yielding 26 megabytes of daily volume from a single sensor channel. Therefore, setting up network communications between an onsite DAQ system and a remote data server is necessary.

Efficient storage of SHM data facilitates easy examination and backup. The best approach for SHM data storage is storing them in a commercial database system or a database storage system from a third party such as MySQL or PostgreSQL. Taking the SHM data files saved in the DAQ system and loaded into the database is necessary (Karbhari and Ansari 2009).

After collecting and analyzing the data from the DAQ system, SHM measurements need to be archived for their protection against data loss and to maintain a chronology record of the structures. Given the high volume of data in the SHM process, downsizing the data for better management is critical. Jeong et al. (2016) used NoSQL database technologies to propose a data management infrastructure framework for bridge monitoring applications.

Koo et al. (2011) presented an SHM data management system (SDMS) based on the MySQL database management system (DBMS) for efficient data storage, retrieval, and sharing of large measurement data sets acquired continuously from SHM systems. However, to solve SHM data management issues, a greater need exists for more efficient tools and integrated systems.

Research Methodology

The proposed framework consists of several modules and submodules. The objective of SHM-based condition assessment module is to assess the building modules' conditions through strain sensor data in an automated manner. Depending on the type of sensors, different methods can be applied. For example, in the case of strain sensors, the threshold value analysis (TVA) method is often employed to detect the exceedance of available threshold strain levels. In contrast, accelerometer data are often processed using different signal processing techniques in time or frequency domains to determine global parameters such as frequencies and mode shapes. By integrating SHM with the BIM model, the detected changes in the system properties or the local damage can be mapped on the BIM model and visualized dynamically. In the present study, only strain sensors are considered for demonstration purposes. Strain sensors can be used to rapidly identify and locate the spot wherein an element exceeds the preset strain threshold. The system is linked to the BIM model to highlight the damaged elements in an automated manner.

By linking SHM to BIM, the efficiency and speed of structural condition assessment processes can be increased, and the process can potentially help nonengineers interact with the building elements and gain an overall sense of the structural condition. Although this system is applicable for all types of structures and infrastructures, its uniqueness in modular construction is in its ability to rapidly detect the buckled or yielded steel members (local damages) in a module, which are often hidden behind the fireproof coating and drywall at different phases (e.g., manufacturing, shipping and installation).

The developed framework consists of four subcomponents, including an Arduino Uno microcontroller (Arduino, Ivrea, Italy) equipped with a strain sensor, an amplifier and Wi-Fi shield (sensory system); Autodesk Revit 2017 (BIM software); Dynamo (visual programming environment); and SQL Server (database management system). In this framework, two links exist: the link between the sensory system and the SQL server and the link between the SQL server and the BIM model.

First, the BIM model is developed with all elements including the virtual sensor and its essential parameters, such as *StrainMaxPoint, DamageFlag, Sensor-ID*, and others. The BIM model is used as a central model to visualize and monitor the strain level remotely produced in critical elements. After developing the BIM model, a specific database is designed in a MySQL environment to house and update the captured sensors data. DAQ systems such as Arduino Uno are coded to remotely send the sensors' measurements in real time to an external database using the Arduino GSM Shield (Arduino, Ivrea, Italy). The Arduino GSM Shield connects the Arduino to the Internet using a GPRS wireless network. Before transferring the sensor data to MySQL server, a schema (database) table and essential parameters such as *Record-ID* and *Sensor-Value* are generated for the strain sensor in the database to accommodate the sensor information and the sampling data. In this study, only strain value is of concern. When strain sensor data are imported into the database, they need to be read by an external tool before processing. For this purpose, a visual programming and computational design tool called Dynamo is utilized for automation purposes (Dynamo BIM 2017).

To link the MySQL database (physical sensors data) and the BIM model (virtual sensors), nine modules were developed and coded in Dynamo to automatically read the strain data stored in the database, sort the data, and update the BIM model with the latest real-time sensor data. These data were used to send notifications to engineers through their wireless devices, such as personal computers or smartphones, enabling them to then take the necessary actions if strain values exceed the predefined strain level. The individual steps of the workflow, developed in Dynamo, are subsequently described.

Once the strain data are read, a module is developed to automatically sort and interpret. For example, when strain sensors are used to measure the real-time strain on critical structural members in the modular system, the developed module sorts the strain data and obtains the maximum strain value in every time interval. Once the strain data are sorted, and the maximum strain value is extracted, an additional module then updates the corresponding virtual sensor parameter in the BIM model. One or more preset strain threshold values can be defined to automatically highlight the structural elements in the BIM model to rapidly identify and locate the elements for which preset threshold strain occurs. In this study, only one threshold is defined. The damaged elements for which the strain exceeds the threshold are highlighted in the BIM model to generate an alarm signal. The hierarchy of the processing modules is shown in Fig. 3.

To illustrate the capabilities of the system developed, a set of strain values is utilized to mimic the data produced by a strain sensor.

System Framework

Conceptual Framework of Wireless Strain Monitoring System

The DAQ system recommended in this framework consists of strain gauges, a microcontroller and its software, an instrumentation amplifier module, a 9V battery, a Wi-Fi shield, jumper wires, and a computer with programmable software.

The system configuration is based on the Arduino Uno open-source microcontroller used for onboard data acquisition. The microcontroller is based on the ATmega328P, an Atmel AVR processor that can be programmed in a computer using the C language through a Universal Serial Bus (USB) port. The microcontroller can sense the environment by receiving input from

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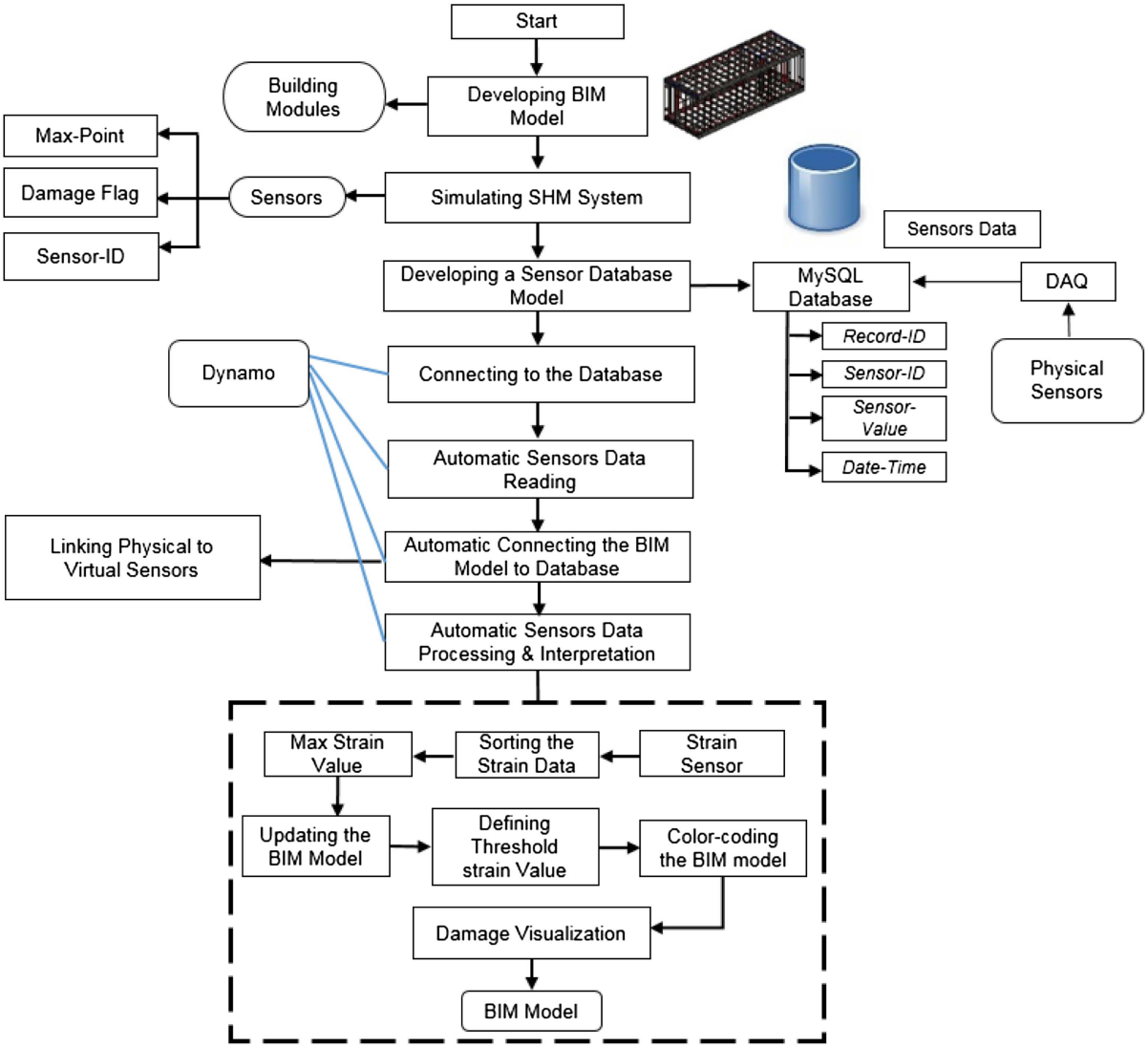


Fig. 3. Hierarchy of steps in this study.

a variety of sensors and can be powered by connecting to a computer with a USB cable, an AC-to-DC adapter, or a battery. In this study, a battery is used during monitoring. The Arduino board can operate on an external supply of 6 to 20 V. However, if supplied with less than 7V, it may be unstable. If using more than 12V, the voltage may overheat and damage the board. Therefore, the range should be between 7 and 12V. Hence, a 9V battery is recommended.

Strain gauges are variable resistors for which the resistance changes when they are stretched or compressed along their length. When bonded to a structural element, the resistance changes proportionally to the strain of the element depending on the force applied to deform it. Although directly reading the voltage change of the bridge of a strain gauge is possible, doing so is normally impractical without an amplifier. The maximum voltage change from the strain is too small for the digital to analog converter in an Arduino to register. The instrumentation amplifier boosts the signal to bring it to a readable range. For this purpose, an instrumentation amplifier, such as the INA125P amplifier, should be used when measuring the strain gauge data with the Arduino.

The open-source Arduino software (IDE) can program the Arduino Uno to process the raw data and extract the results of interest to be transmitted to a central station, thus reducing the data communication demand that is particularly suitable for wireless communication in SHM systems. Storing a sensor's data over a cloud-based database server such as an SQL database is also possible. Creating a website or another app to read the sensor values in the SQL database and track the events according to the threshold values will give considerable power to any SHM project. For this purpose, a Wi-Fi shield compatible with the Arduino Uno is recommended. An Arduino compatible shield, such as the ESP8266 WiFi Shield, equips the Arduino with the ability to connect to the wireless networks that can be used for Internet of Things (IoT) or WiFi-related projects. Monitoring data such as strain values can be stored directly in a MySQL server using the MySQL Connector in the Arduino library. The MySQL Connector in the Arduino library can be used to connect the Arduino project directly to a MySQL server without using an intermediate computer or a web- or cloud-based service. When the strain values are measured and stored in a prebuilt MySQL database, they can be analyzed in remote wireless connected devices such as personal computers, tablets, or smartphones. An alarm system could be proposed in these devices for rapid detection of excessive deformation (plastic deformation) in structural elements for timely repair or replacement of the faulty elements. Fig. 4 illustrates the

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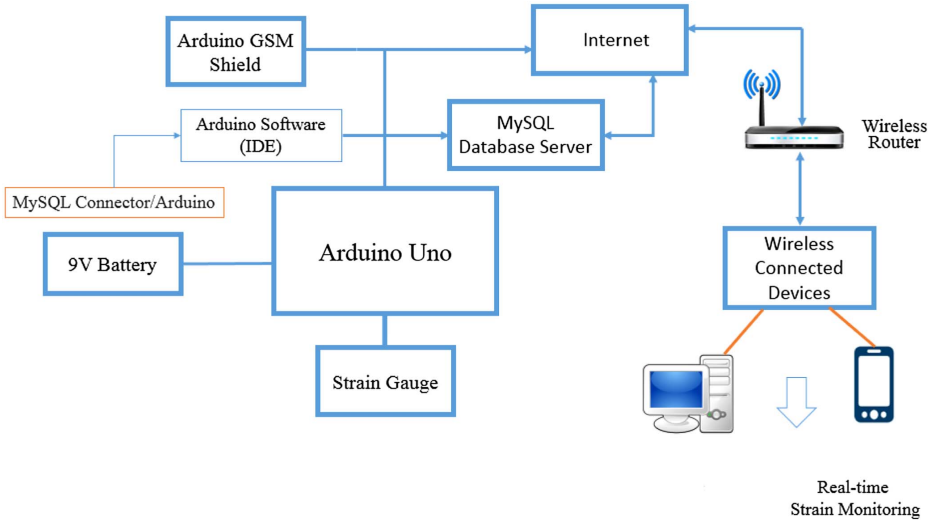


Fig. 4. Conceptual framework of wireless strain monitoring system.

architecture of a conceptual framework of a wireless strain monitoring system.

SHM System Modeling in the BIM Model

After developing the BIM model, SHM sensors need to be simulated in the model. A strain sensor is modeled and attached to its specified locations in the BIM model. Many different categories exist in the BIM objects, such as column, beam, floor, roof, door, window, and others. In BIM, each category has its own IFC class name, such as *IfcColumn* and *IfcRoof* to name a few, and in sensors—*IfcSensor* and *IfcSensorType* class names are categorized under the specific equipment category. Therefore, sensors in the BIM model must be developed subject to the specialty equipment category.

Different parameters are defined for the strain sensor during modeling: identity data (e.g., manufacturer, label, mark, model, cost), phasing, scope, station, and data (StrainMaxPoint, DamageFlag). Station parameter is defined to show the exact location of the sensors in the BIM model, and the mark parameter is set to link the physical sensors to virtual sensors in the BIM model. In fact, the physical sensors' specific identification (ID) must be assigned to each corresponding virtual sensor in the BIM model to link the physical and virtual sensor. StrainMaxPoint is the parameter created to accommodate the maximum strain value recorded by the strain sensor at each time interval, and DamageFlag is generated to show the structural elements condition rate. A building module and a strain sensor modeled in the BIM tool, along with some of its defined parameters, are illustrated in Fig. 5.

Defining the Sensor Database Model

To insert the sensed data from a sensor into a database such as MySQL, a schema (database) and a table, along with all essential parameters, are to be defined. The table will embody the sensor data received from the DAQ system. To prevent users from manually generating the sensor database model, two modules are created in Dynamo to draw a schema, producing a table and columns in an automated manner. The schema—shmsystem—is defined along with a table called *strainsensor_1*. Three parameters are introduced for the *strainsensor_1* table: *Record_ID, Sensor_Value, and Sensor_Name*, where *Record_ID* constitutes the primary key. Appropriate nodes, code blocks, and connections are required to automate this process. The modules developed to generate the schema and table in the MySQL database in an automatic sense and to embody the strain sensor measurements are illustrated in Fig. 6.

Dynamo supports the use of code blocks, elements containing small scripts written in a textual programming language such as C++ and Python. These code blocks allow for the generation of short algorithms that introduce more complex functionalities that are not possible to be generated by other nodes. As shown in Fig. 6, two code blocks are applied when generating a schema and its table, and one code block is employed to create the parameters of the table.

The database model can be expanded in the case of using strain sensors for multiple structural elements, and if a comprehensive database model for the entire sensory system, the information on the corresponding structural elements and their condition are required. Fig. 7 shows the entity relationship diagram (ERD) of the proposed database model when using multiple strain sensors (e.g., four strain sensors) in the system.

As shown in Fig. 7, the ERD consists of six entities: *Strainsensor_1*, *Strainsensor_2*, *Strainsensor_3*, *Strainsensor_4*, *structural_element*, and *condition*. The relationship among the entities shows that a one-to-many relationship exists between *strainsensor* and *structural_element* entities, and a one-to-one relationship between *structural_element* and *condition* entities. Therefore, each strain sensor is installed on only one structural element, but a structural element may have multiple sensors. Each structural element may also have a condition: undamaged, partially damaged, or severely damaged.

Connecting to the Database

After generating modules for developing a sensor database model, the module(s) need to be connected to another module(s) to be integrated into the database. Consequently, a new module needs to be generated for connecting other modules to the MySQL database. In this module, several code blocks are involved in inserting server and port number, user id, and password. In fact, this module performs as a central module to which all other modules are to be connected.

Extracting Parameters from the BIM Model

Here, a module is generated to retrieve the virtual sensor parameters from the BIM model. The sensor parameters need to be obtained to check the latest values of these parameters in the BIM model.

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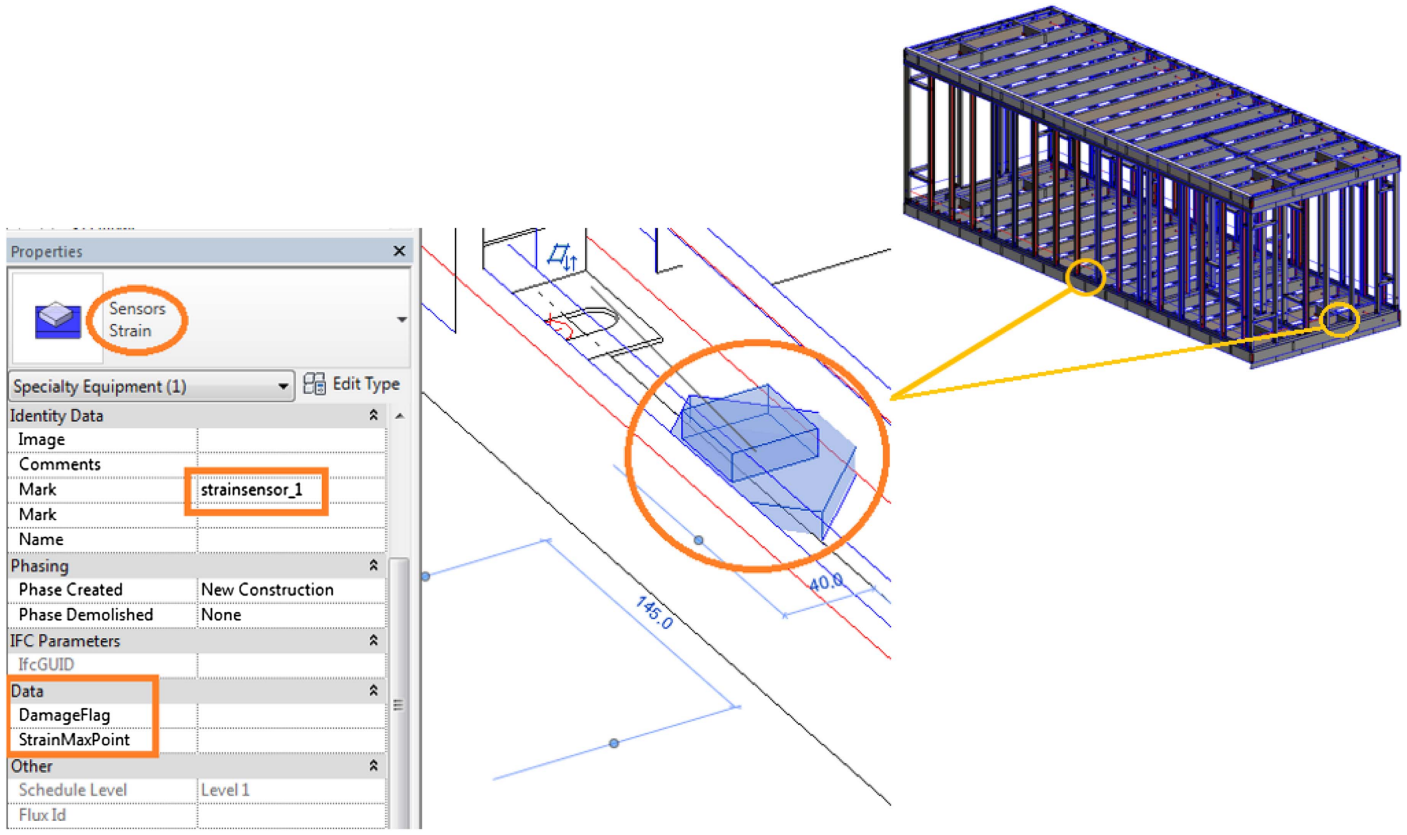


Fig. 5. Developing BIM model and virtual strain sensor.

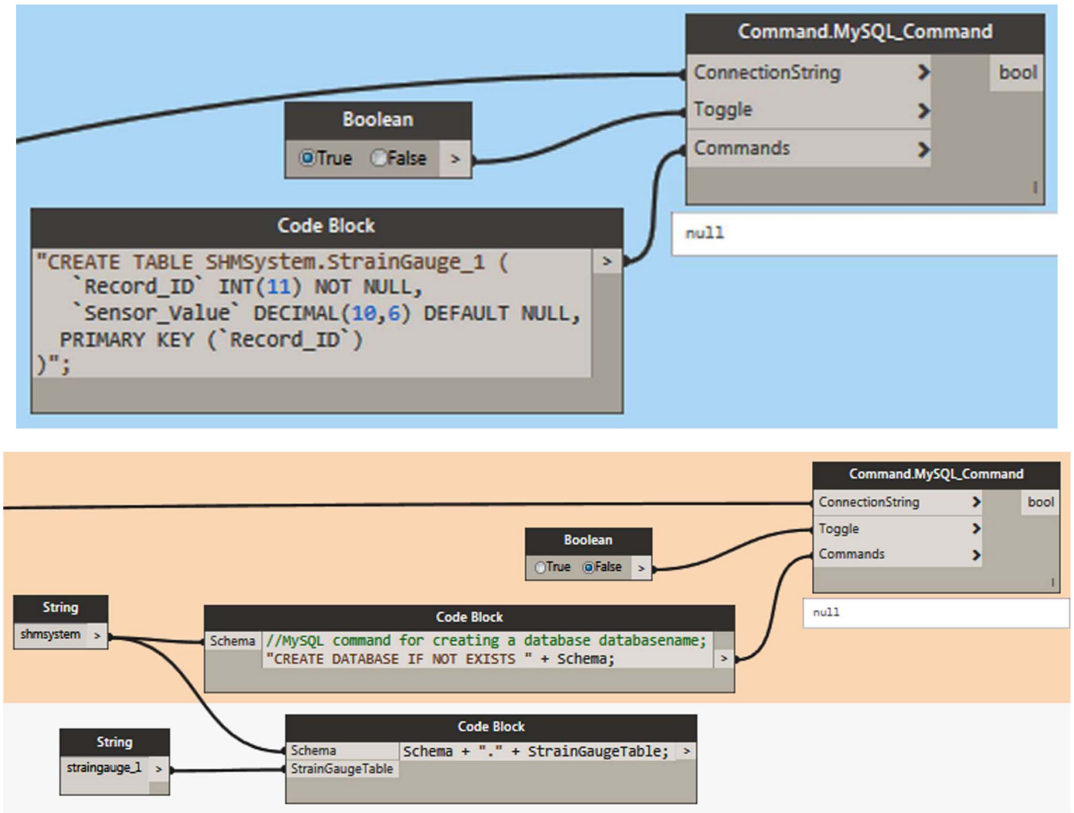


Fig. 6. Automated generation of schema and table for strain gauge in database.

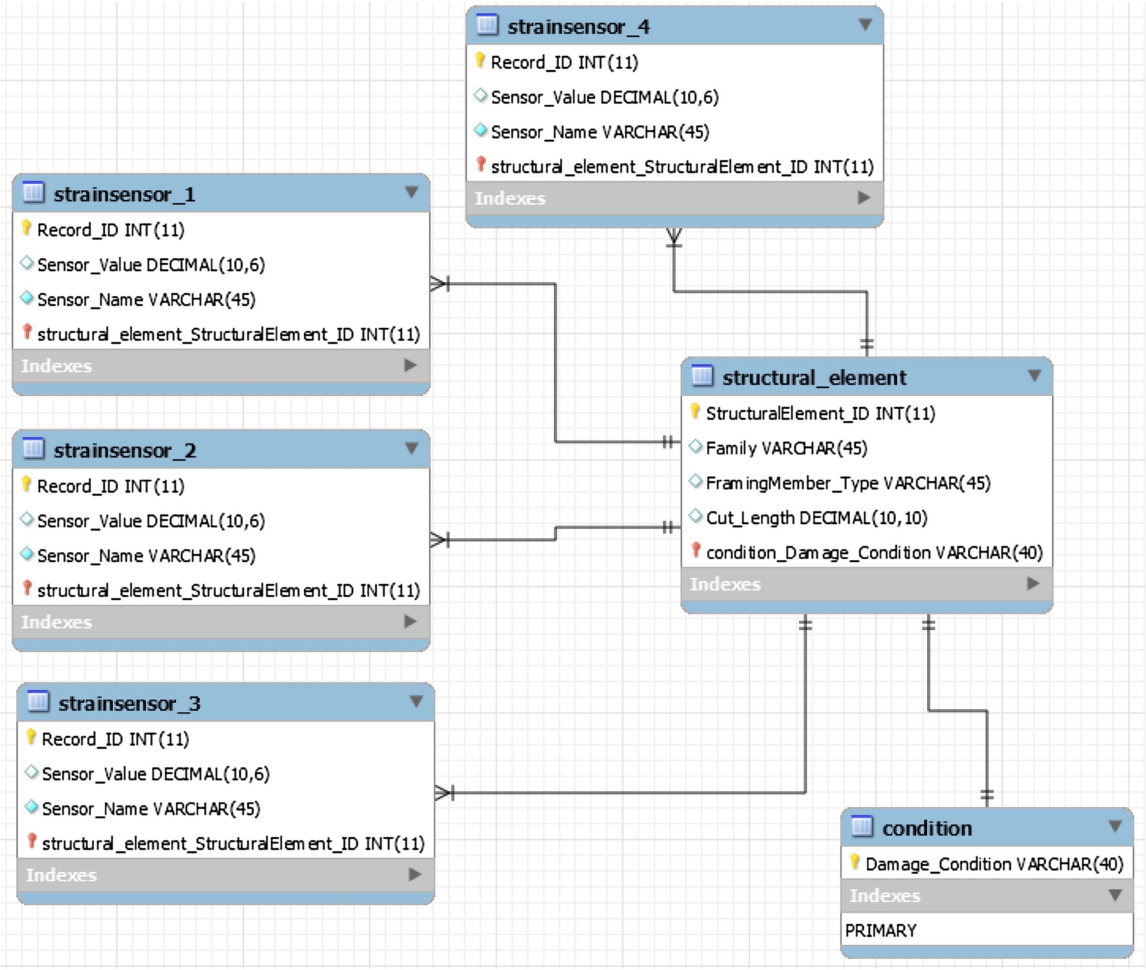


Fig. 7. ERD of SHM system database.

In this module, first, the specialty equipment category is read from the list of categories in the BIM model because the virtual strain sensor object is modeled in this category. Next, all elements are selected in this category (virtual sensors in the BIM model). Then, the user-defined sensor parameters such as *StrainMaxPoint* and *DamageFlag*, the values of which are to be displayed, are extracted from the BIM model and shown in one list.

BIM Model to the Database Linkage

After extracting the essential parameters from the BIM model, they need to be linked to the parameters in the database already generated for each sensor. This integration is required for any future BIM model updating. A module is developed to connect the parameters extracted from the BIM model to the sensor parameters defined in the MySQL database.

Using this module, the *Sensor_Value* parameter generated in the MySQL database is linked to the *StrainMaxPoint* parameter generated in the BIM model. Inevitably, the module needs to be connected to the modules previously developed.

Automatic Reading of Sensor Values from MySQL Database Server

When a connection was established between the BIM model and the MySQL database server, the next step is to automatically read the sensed data from the MySQL database.

Different nodes are applied and connected to one another for this automatic sensor data reading. As previously mentioned, the *StrainMaxPoint* parameter is generated and assigned to the virtual sensor in the BIM model. Consequently, only the maximum strain value at every time interval is to be sent to the BIM model to update the *StrainMaxPoint* parameter of the virtual strain sensor. In this module, to find the maximum value of the strain at each time interval, the strain values are sorted because, for comparison purposes, only the maximum volume of the strain is required. If the maximum strain value is less than the predefined threshold strain value, then the structural element is not damaged and remains intact.

Updating the Virtual Sensor Parameter in the BIM Model

After retrieving and sorting the sensed values stored in the MySQL database, the associated virtual sensor parameters are updated in the BIM model using a module developed for that purpose.

In this module, the *SetParamaterByName* node is used to update the *StrainMaxPoint* parameter of the strain sensor in the BIM model based on the maximum strain value recorded from the MySQL database. The maximum strain value is recorded at the end of each time interval, and the corresponding parameter in the BIM model is updated accordingly. A set of modules is developed for defining the threshold strain values and color coding schemes for the damaged structural components to represent their

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status in the BIM model. Therefore, an alarm system can be generated and transmitted to the responsible personnel to attend to the structure.

Defining the Threshold Strain Value

Identifying an appropriate threshold value is essential for strain monitoring of critical structural elements. The allowable strain in the studs can be determined based the yield strength (207.0 MPa) and Elastic Modulus (210,000 MPa) of the material, which is around 990 microstrain ($\mu\varepsilon$) for the present case.

By applying this module, the condition of the *DamageFlag* parameter of the virtual strain sensor is classified into the following two cases: undamaged and damaged. If the maximum strain measurement does not exceed the predefined threshold strain value, the *DamageFlag* parameter is considered undamaged. Otherwise, it is flagged as damaged. To check these conditions, this module needs to be connected to the module previously developed for which the *StrainMaxPoint* parameter value is applied as one of the inputs that indicates the maximum strain value measured by the physical strain sensor.

Mapping a Virtual Sensor in the BIM Model to the Intersecting Structural Framing Members

To visualize damaged elements in the BIM model, a parameter called *condition* needs to be updated based on the strain sensor's *DamageFlag* parameter. If the *DamageFlag* parameter is set, the structural element's *condition* parameter indicates the member as damaged. Before upgrading the *condition* parameter based on the *DamageFlag* parameter, the virtual sensor intersection by the correlating structural framing member in the BIM model must be assured.

In the module performing this function, first, the geometry-related intersection between the virtual sensors (specialty equipment category) and structural members (structural framing category) is checked. If an intersection is found between these two members, then the module will update the *condition* parameter of the structural member based on the *DamageFlag* parameter of the correlating sensor in the BIM model. Accordingly, the damaged structural members are identified if their *condition* parameter identifies the member as damaged.

Color-Coding of Damage Structural BIM Element

After upgrading the 'Condition' parameter for structural framing members, the damaged elements are highlighted using a predefined color code. It works as an alarm system to highlight the elements where the strain produced by external forces exceed the pre-defined threshold value. For this purpose, a module is created, in which, the structural framing elements with damaged 'condition' parameter are filtered from the list of all structural framing members and highlighted in the BIM model.

Model Implementation

The initial step of integrating BIM into the SHM process is to link the physical sensors to virtual sensors and connect them to real data from the field. This integration is possible by attaching the physical sensor ID to the virtual sensor in the BIM model. To do so, first, a BIM model of an individual four-sided steel module is developed. Second, the process modules, as described in the previous section, are generated to introduce a workflow to link the virtual to the physical strain sensor, update the associated parameters, such as

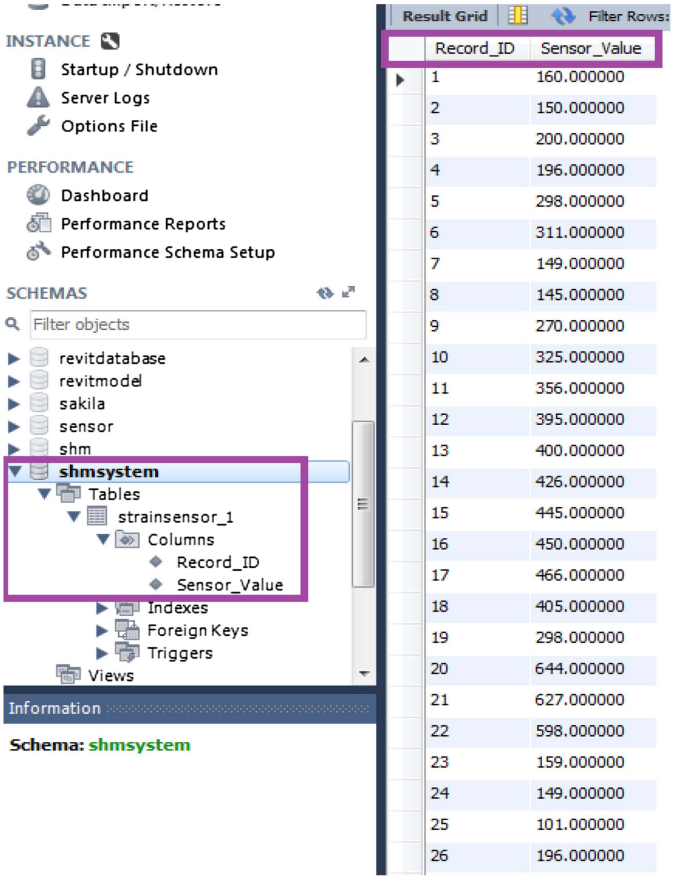


Fig. 8. Physical strain sensor reading introduced to database.

StrainMaxPoint, *DamageFlag*, and *condition*, and then highlight the damaged structural framing members in the BIM model.

The Arduino-based DAQ system can be coded to transfer the sensed data to an external database such as the MySQL database server. Before sending and accommodating the sensor data, the schema, tables, and relevant parameters need to be generated in the MySQL database. When sensors' data are sent to the database, they should be preprocessed before becoming exposed to further analysis. In this study, a list of one hundred strain data points is introduced to a MySQL database. The strain values and their record IDs are imported into the MySQL database, as shown in Fig. 8.

As observed in Fig. 8, the *shmsystem* schema and the *strainsensor_1* table with three parameters, including *Record_ID*, *Sensor_Value*, and *Sensor_Name*, are generated in the MySQL database before introducing the strain data. Once the strain data are added to the database, the values are read from the MySQL database server and sorted through the module previously developed in an automatic sense. All parameters are captured, read, and sorted. The maximum strain reading from the list of strain values is extracted after the values are sorted because, if the maximum strain value of an element in every time interval does not exceed the allowable strain (threshold) value, the element is regarded as undamaged. In contrast, if the maximum strain produced in a structural member exceeds the predefined allowable strain value, then the element is damaged and needs specific considerations and corrective action. How the strain sensor values are read from the predefined schema and sorted to update the *StrainMaxPoint* parameter of the virtual strain sensor in the BIM model is illustrated in Fig. 9.

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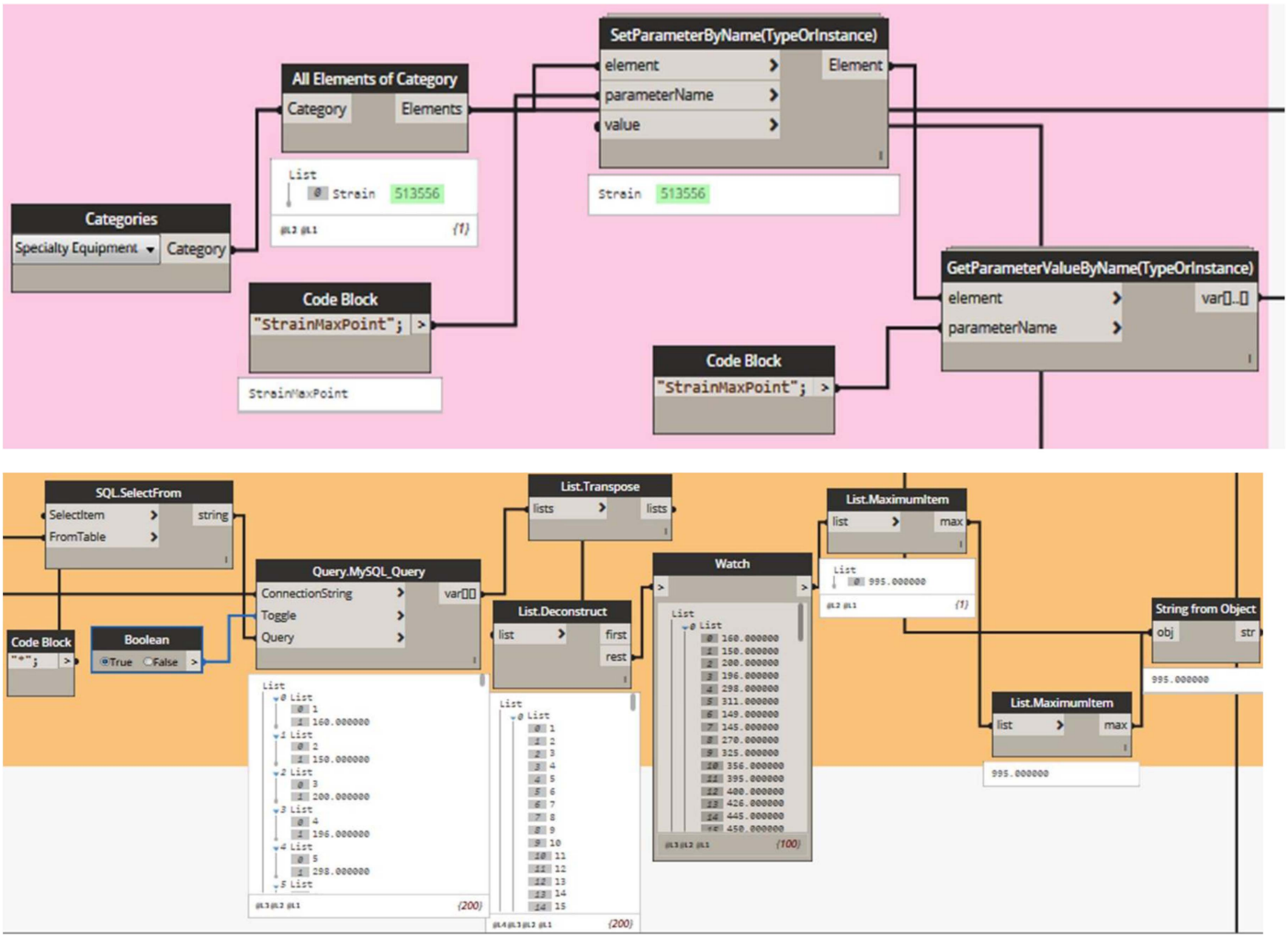


Fig. 9. Automatic reading of strain sensor data and updating BIM model.

As observed in Fig. 9, only one strain sensor with ID number 513556 exists in the example BIM model. After reading the strain values from the database, the maximum strain value is found to be 995 $\mu\varepsilon$, which is transferred to the next module to update the *StrainMaxPoint* parameter of the corresponding virtual sensor in the BIM model. The virtual strain sensor and its parameter updated in the BIM model are shown in Fig. 10.

As observed in Fig. 10, after running the program, the *StrainMaxPoint* parameter is automatically updated. This step is an initial damage visualization process in the BIM model in a sense that the damage detection scenario begins from transferring the *DamageFlag* parameter information from a virtual strain sensor to the correlating structural framing member. By adopting the previously described module, the *DamageFlag* parameter becomes *damaged* because the extracted maximum strain value of 995 $\mu\varepsilon$ is more than the predefined threshold strain value of 990 $\mu\varepsilon$. To upgrade the *condition* parameter of the monitored structural member based on the *DamageFlag* parameter of the virtual strain sensor, the intersection between the virtual sensor and correlating structural framing member needs to be checked in the BIM model through the previously described module, as shown in Fig. 11.

First, the geometry of all structural framing elements and virtual sensors are extracted from the BIM model. To identify the structural members to which the virtual sensors are attached, the structural members are filtered to separate the structural member to which the strain sensor is connected from other members. When the correlating structural member is separated from others, its *condition* parameter is updated based on the *DamageFlag* parameter of the corresponding virtual strain sensor. Consequently, because the *DamageFlag* parameter in the virtual strain sensor indicates the damage state, the *condition* parameter is updated to the damage state in the correlating structural member. When the *condition* parameter in a structural member indicates the damage state, the damaged element is highlighted in the BIM model through the module as described previously. As shown in Fig. 12, the structural member to which the virtual strain sensor is attached is damaged and is highlighted in the BIM model shown in Fig. 13.

One of the advantages directly attributed to BIM in the SHM integration process is that all of the information of the structure generated during its lifecycle is accessible, which can facilitate easier decision making for repair and maintenance plans. For instance, information on a damaged member such as material, type, cut length, type of connections, manufacturer address, phone number, date, time, minimum number of workers needed, and others, can all be easily extracted from the BIM model. The architecture of the developed system workflow and the correlation among its modules are shown in Fig. 14. If the modules are not correctly connected to each other, the system might not work.

In Fig. 14, the modules were numbered to better demonstrate their linkage. In Fig. 14, the output of module number 1 that links the system to the MySQL server is connected to the following

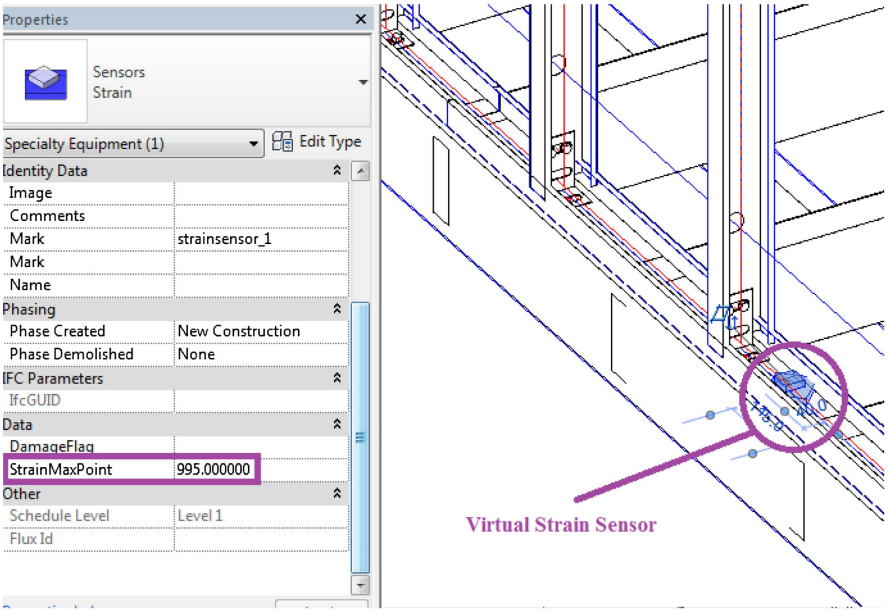


Fig. 10. Updating BIM parameter based on maximum strain value in $\mu\varepsilon$.

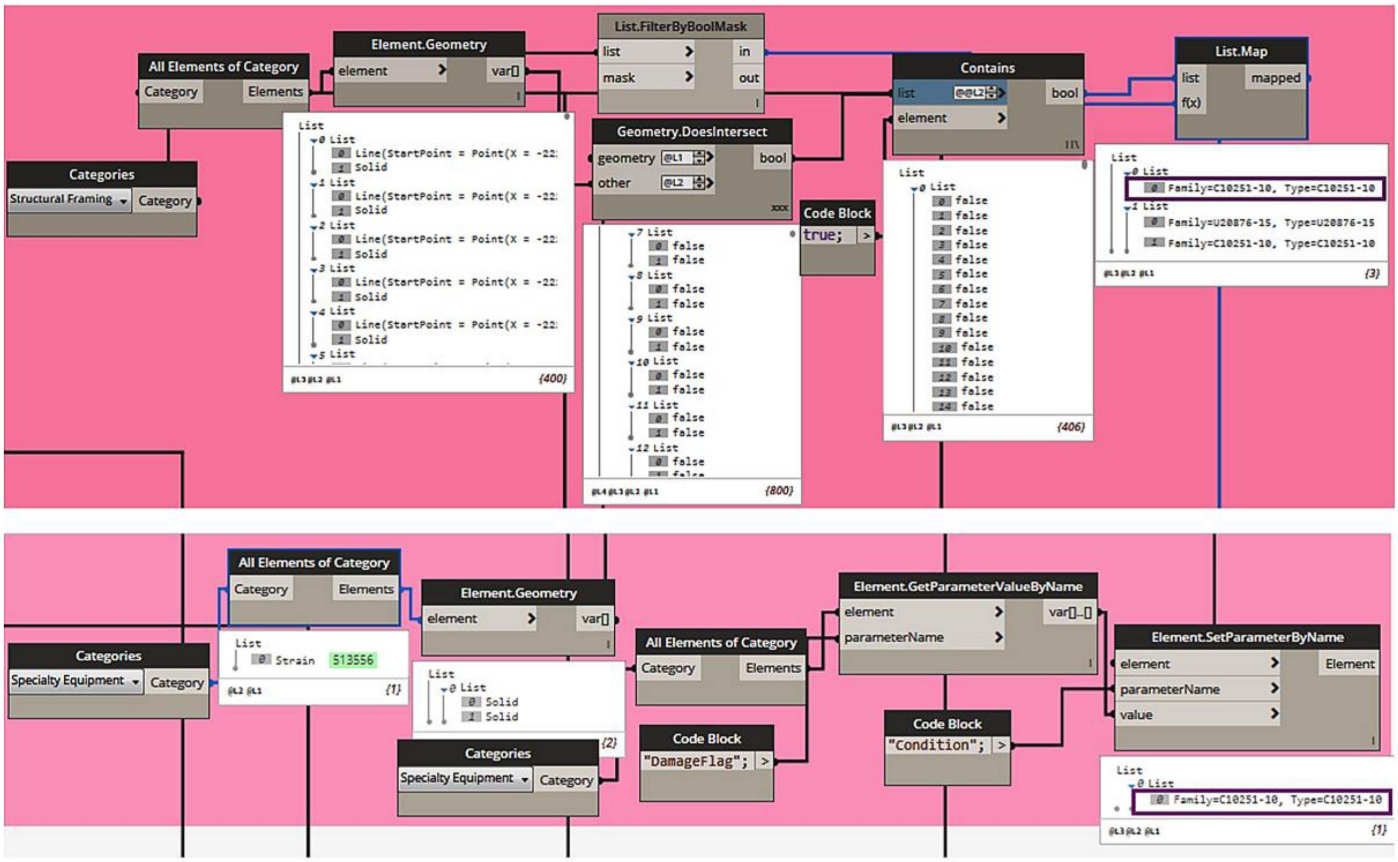


Fig. 11. Checking intersection between virtual sensor and BIM structural element.

modules: number 2 (to connect the system to the monitoring database and table in the database), number 4 (to link the BIM model to the database containing sensor data), and module number 5 (to retrieve the sensor data from the database). The output of module number 3 (to extract strain sensor parameters from the BIM model) is connected to module number 4. As shown in Fig. 14, a link exists between the following modules: numbers 2 and 4 to connect the maximum strain value in each event to the corresponding parameter in the BIM model. The output of module number 2 is connected to module number 5 (to automatically read and sort strain values

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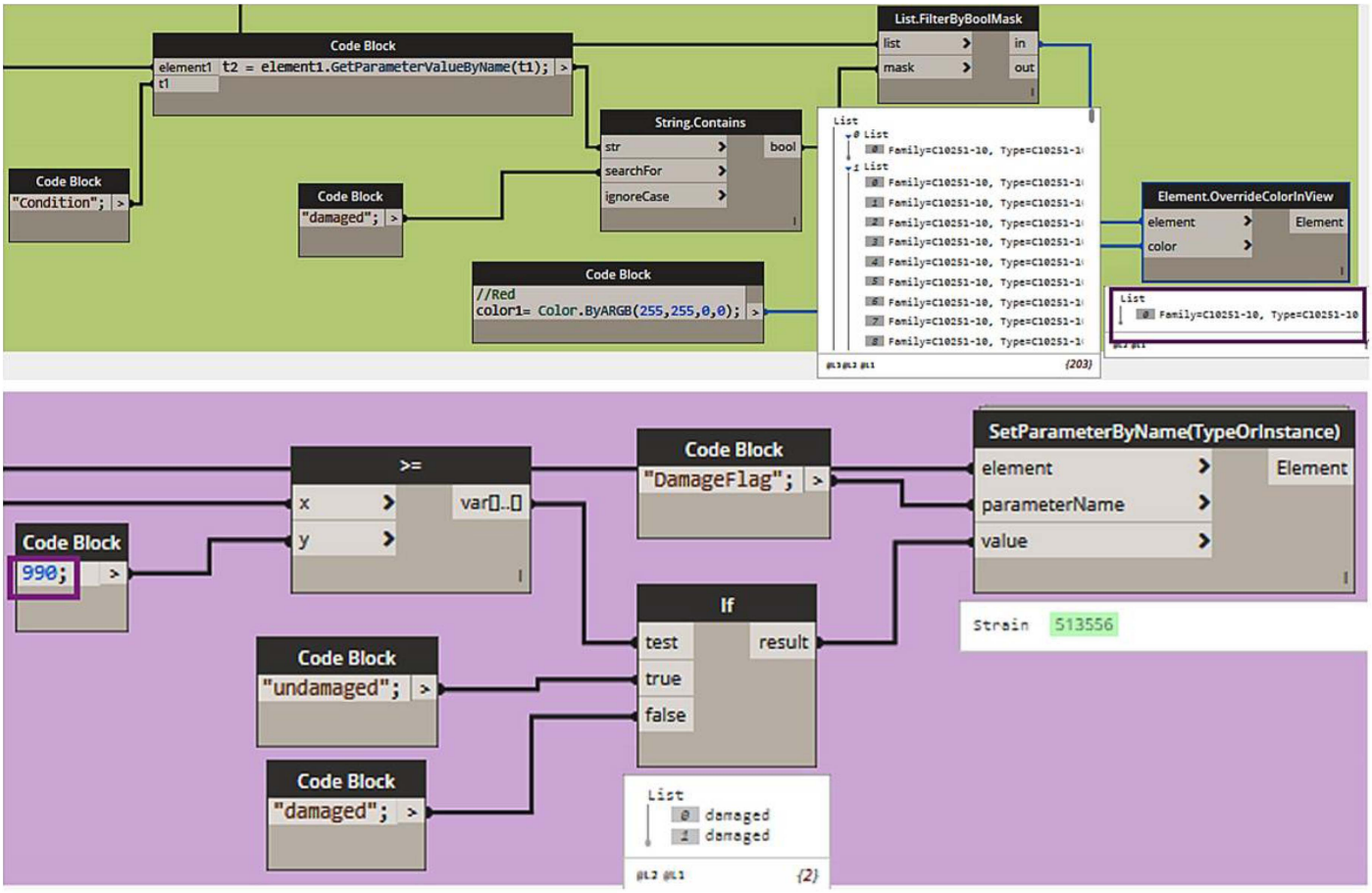


Fig. 12. Threshold strain value definition and color-coding of damaged structural BIM elements.

stored in the database). The output of module number 5 is connected to module number 6 to update the sensor parameters in the BIM model. Finally, to automatically update and highlight the affected structural element in the BIM model based the threshold strain value defined in module number 8, module number 7 is connected to modules number 6 and 9 in the developed workflow.

Discussion

The main purpose of this paper was to investigate the potential of BIM in the damage visualization and rapid detection of damaged elements in structures such as building modules. This paper introduced a novel integrated strain-based monitoring system framework in BIM. The small size and low cost of the conceptual sensory system proposed in this study can be effective in modular structures that usually consist of small and narrow components. The system can also be utilized for rapid condition assessment of structural components. For this purpose, monitoring data were integrated into BIM through a novel workflow developed in a visual programming environment. The corresponding structural elements are automatically highlighted in the BIM model when strain measurements surpass the predefined threshold. Exceeding the strain threshold can be considered as damage or unacceptable deformations that can be further accentuated by repeated loading, especially during the operational phase.

To show the capabilities of the developed system, a set of strain values stored in a MySQL database was utilized to mimic the data produced by the proposed wireless strain gauge sensory system. Additionally, the BIM model of an individual light gauge steel module was used to demonstrate the capability of the developed framework in terms of damage visualization. The strain values studied in this paper were analyzed and compared with the threshold values detected by the developed system. For automatic recording, sorting and updating of strain values, and linking between the physical and virtual sensors (in the BIM model), a novel workflow consisting of nine modules as shown in Fig. 13 was developed in a visual programming environment called Dynamo. The developed system was able to successfully transfer the sensor data stored in a central database server and map the strain sensors to the corresponding structural elements in the BIM model and update the related structural element parameters. The system was able to detect the maximum strain values for each event as well as the values beyond the predefined threshold values. Through a color-coding scheme, the system was also able to automatically highlight the structural element in which case the strain values surpassed the threshold values. Incorporating BIM into the strain monitoring system provides engineers and facility managers the additional information required for deciding on repair and maintenance.

Although the approach is quite general and could be applied to any type of structure, such as buildings and bridges, this paper focuses on application to modular buildings as a case study. Additionally, modular construction has some challenges that differ from conventionally constructed buildings and bridges. In this case, the modules are subjected to different types of loads, such as manufacturing, transportation, handling, installation, and operational loads during their life cycle, and the main challenge is to detect buckled or yielded steel members (local damages)

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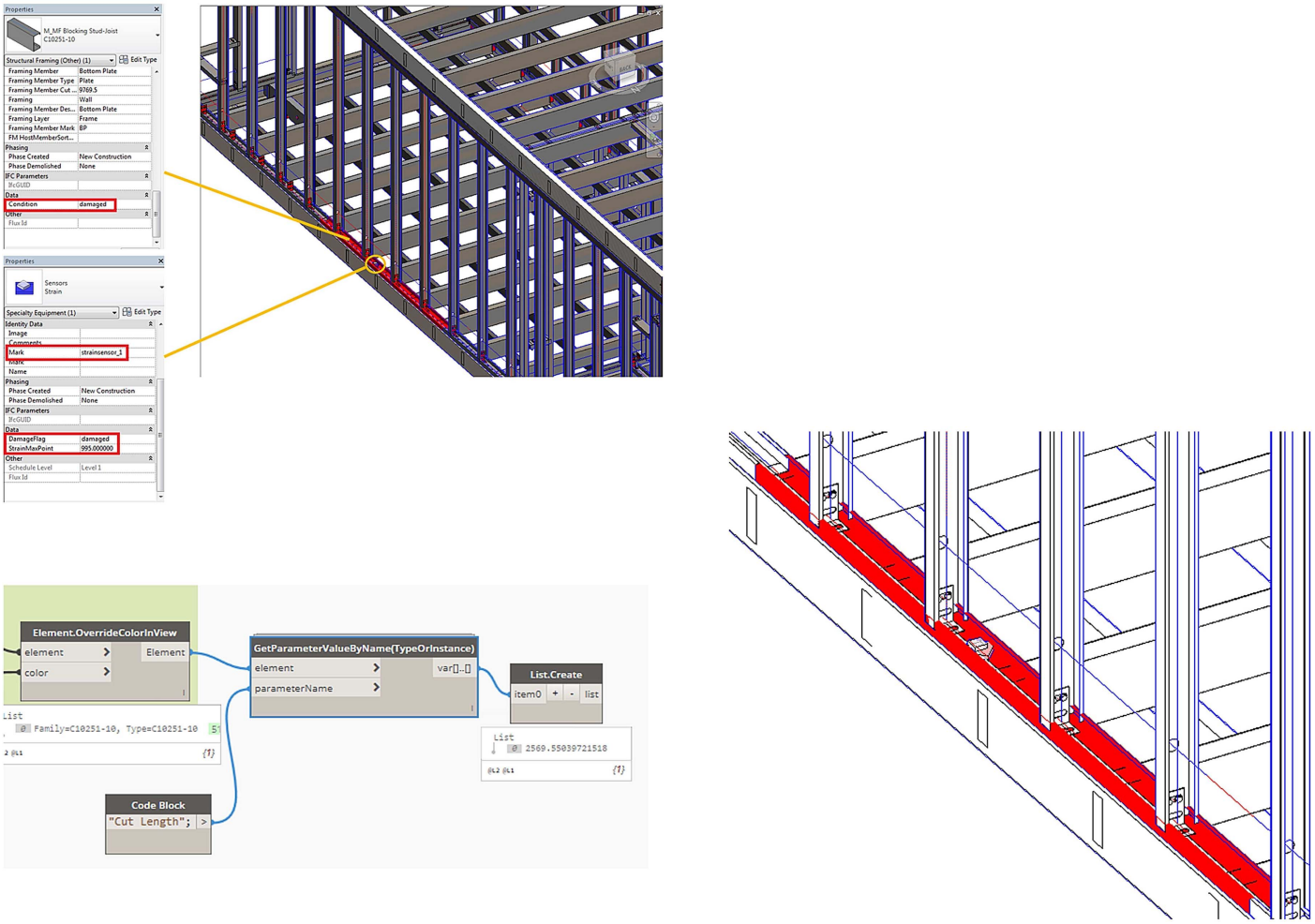


Fig. 13. Three-dimensional damage visualization of damaged structural BIM element.

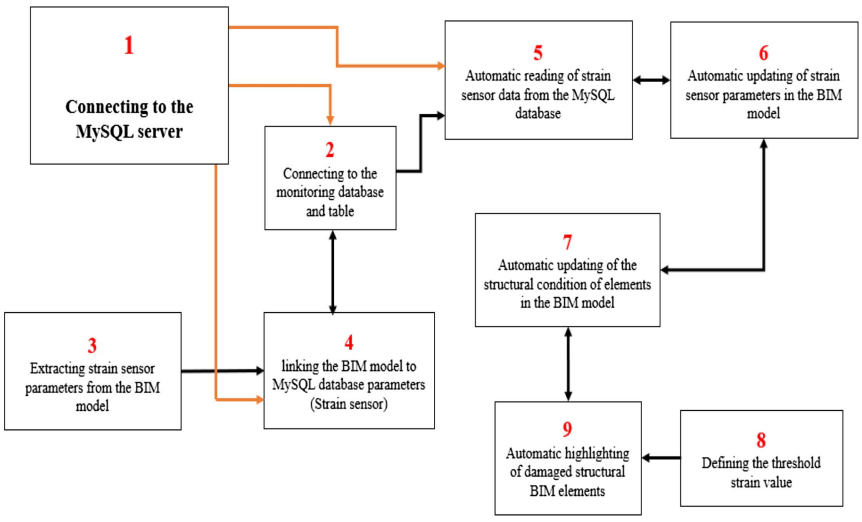


Fig. 14. Workflow for integrating BIM into SHM process.

in a module, often hidden behind fireproof coating and drywall. For example, any possible damage that occurs in the building modules during transportation may cause miss-alignment issues during installation, affecting the structural performance and disrupting continuity in the envelope, which may significantly increase air leakage. Manufacturers usually buy shipping insurance that covers the all-risk and basic risk conditions to be reimbursed for repair costs of the modules if they are damaged during

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transportation. Therefore, the developed system can be used as an efficient, reliable, and rapid monitoring technique for timely detection of damages in modules after transportation and when claiming the repair and modification costs from the insurance company. The system is crucial to monitor the state of the critical elements of the modules in each phase to ensure their acceptable conditions before going to the next phase.

However, this work has some limitations, which are as follows.

1. The system developed here was tested using a limited number of sensors, and only strain sensors were used. The system should be tested further with a larger number and multiple types of sensors. Data from different types of sensors are interpreted in different ways. For example, strain data can be compared to a threshold value for a certain level of damage, whereas acceleration data can be used to determine the change in vibration modes and frequencies.
2. Deformation is a good parameter for localized damage detection, but a very good understanding of the structure is needed to design the instrumentation plan. If several similar types of members/connections are present and subjected to the same order of forces, identifying the most critical monitoring elements may not be trivial, and installing one or more sensor(s) on each element is not cost effective. Further study is required to automatically identify the critical locations and sensors.

Conclusions

Utilizing an effective data management system, such as BIM, for buildings can be extremely helpful for a rapid and efficient structural condition assessment and visualization. As a comprehensive, detailed, information-based strategy, BIM provides all of the information about the damaged elements, which is useful for effective repair/replacement of the damaged elements. Because the hidden structural damages (e.g., excessive deformation of structural elements in building modules) occurred during transportation or handling processes, they may lead to misalignment risks during installation. Rapid detection of such damages and repair or replacement of the damaged components can help mitigate the misalignment risks and the possibility of further damage during installation.

An SHM framework in which a BIM was adopted for automated and graphical structural condition monitoring, particularly for modular structures, was proposed in this article. Building modules are more susceptible to damage during transportation and installation. In such cases, the structural elements are often hidden, and structural damage such as excessive deformation is difficult to detect. BIM can serve as an ideal four-dimensional graphical computing environment for SHM and repair and maintenance plans. The ability of BIM to visualize the SHM information of a structure will be beneficial for engineers and decision makers because it provides accurate and updated information on the current state of structural elements, structural reliability, and maintenance work to be scheduled. A BIM-based integrated structural health framework was developed for rapid and efficient monitoring of structural deformation. Based on the findings, the following observations were made.

1. A virtual sensor's parameter in the BIM model can be updated based on the measured data from the corresponding physical sensor. This feature was applied to determine when a predefined threshold strain level is exceeded at the monitored location of a structure.
2. The capability of the developed framework was examined through a set of strain values stored in a specially designed MySQL database and a BIM model.
3. The state of a concerned structural framing member in the BIM model was updated and highlighted when the strain value exceeded the threshold value. Data from other sensors, such as an accelerometer, or temperature can be integrated in the same way.

Although the present study demonstrates the feasibility of remote strain sensing and decision making to monitor the state of a hidden structural element in a building module during its life cycle, further studies are required to conduct a large-scale investigation that uses multiple types of sensors to monitor the critical elements with different types of materials in a module. Additionally, a need exists for experimental validation of the developed system using real-life applications.

Acknowledgments

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